Nontermination on a Random Tape

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Abstract

An algorithm may terminate almost surely and still run forever on exceptional random tapes. This paper studies three features of this exceptional set: the survival tail, the Kolmogorov complexity of an individual nonterminating tape, and the Hausdorff dimension of the entire set. For variable-model resampling, reading a finite run backward yields an exact likelihood identity: the information revealed by the repaired events, minus the information needed to encode their order, is a record-to-tape log-likelihood ratio, denoted by Δ_T. Consequently, $\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}$; under computability, the same ratio gives a $\Delta_T - O(1)$ bit saving when describing every surviving tape prefix. Running time and pointwise complexity are therefore controlled by the same likelihood ratio.

For rejection sampling, the survival law and both dimensions are computed exactly. A fair random walk killed at zero exhibits a different infinite-state regime: it terminates almost surely with survival probability of order $T^{-1/2}$, yet its nonterminating set has Hausdorff dimension one and contains tapes of effective strong dimension one. Two whole-block systems with the same event–variable incidence graph, event probabilities, and all pairwise intersection probabilities nevertheless have different survival exponents and dimensions. Finally, unrestricted countable-state machines realize every effectively closed subset of Cantor space. Exact nontermination geometry therefore requires more information than dependency graphs provide and more structure than countable state alone.

1 Introduction

A randomized computation can terminate with probability one and still run forever on a rare set of random tapes. This paper asks three questions about this exceptional set: how rapidly the probability of prolonged execution decays, how much algorithmic randomness one nonterminating tape can retain, and how large the set of all nonterminating tapes is.

Once the tape is fixed, the computation is deterministic. Let $\mathbb{P}$ denote the source law of the tape and let

$$\mathcal{N} = \{\omega : \tau(\omega) = \infty\},$$

where τ is the number of steps before stopping. The tail $\mathbb{P}\{\tau \geq T\}$ measures how often the computation survives for T steps. When the following extended limit exists, its base-two survival exponent is

$$\alpha = -\lim_{T\to\infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

This is a distributional quantity. Kolmogorov complexity instead concerns one tape, and Hausdorff dimension concerns the geometry of the whole set $\mathcal{N}$. A null set may nevertheless have large Hausdorff dimension and contain tapes of high effective dimension.

For resampling algorithms, the link begins with a finite-run identity. Suppose that a run repairs events $A_1, \ldots, A_T$, with marginal probabilities p_{A_t}, and let Q_T be a subprobability law used to describe the ordered repair history. Reading the run backward gives

$$\Delta_T = \sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(A_1, \ldots, A_T)} = \log_2 \frac{\text{backward record mass}}{\text{source tape mass}}. \tag{1}$$

The first term is the marginal information in the repaired events; the second pays for their order. Because the numerator has total mass at most one,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \tag{2}$$

More precisely, Corollary 3.3 decomposes the survival information $\log_2(1/\mathbb{P}(E_T))$ into the conditional mean of Δ_T, a relative-entropy mismatch, and the unused mass of the backward law. When the laws are computable, coding the same backward mass shows that Δ_T source bits, up to an additive constant, can be saved in a description of each tape prefix that produces the run. The exact factorization in (1), rather than the elementary likelihood-ratio bound (2), is the resampling-specific part of the argument. A consistent comparison law on infinite surviving tapes is the additional ingredient that turns these prefix bounds into a Hausdorff dimension formula.

Two examples show the range of possible behavior. Rejection sampling is the simplest sharp exponential case: survival is exponential, and one tilted source law gives both the largest effective dimension of an individual nonterminating tape and the Hausdorff dimension of all such tapes. A fair random walk killed at zero is the complementary countable-state case. It stops almost surely with survival probability of order $T^{-1/2}$, although its nonterminating tapes form a null set of Hausdorff dimension one and include tapes of effective dimension one. Thus an exponential rate and an asymptotic dimension can both miss a nontrivial logarithmic cost of survival.

The Lovász local lemma has a different scope. It gives conditions under which bad events can be avoided, and its algorithmic forms give termination and tail guarantees from event probabilities and dependency structure (Erdős and Lovász, 1975; Moser and Tardos, 2010; Haeupler and Harris, 2017). Here a particular computation is fixed and the aim is to determine its exact survival law and its exceptional tapes. Two whole-block systems constructed below have the same event–variable incidence graph, event probabilities, and all pairwise intersection probabilities, but different survival exponents and dimensions. This is not a stronger local lemma; it shows that the usual graph data do not identify these exact quantities for a fixed process.

The ingredients used below are classical separately: effective dimension, source-relative Hausdorff dimension, finite-state pressure, and reversible encodings in entropy compression (Billingsley, 1965; Athreya et al., 2007; Simpson, 2015; Moser, 2009; Moser and Tardos, 2010). The paper does not introduce a new notion of dimension. Its main result is the backward resampling factorization and the tail and Kolmogorov-complexity bounds obtained from it. The matched whole-block pair proves that even incidence-graph, marginal, and pairwise intersection data do not determine the exact survival exponent, maximal effective strong dimension, or Hausdorff dimension. The finite-state formulas and the probability calculations used as benchmarks are stated with their classical provenance.

The pointwise question—what can be said about one realized object rather than only an average or a class—is motivated in part by recent work on individual trained networks and Gaussian fields (Li and Xu, 2026; Xu, 2026b). The search for a smaller state that still determines a sequential computation is related in spirit to Bellman-sufficient information (Xu, 2026a). These connections are motivational; none is used in the proofs.

2 Tape dimension and rejection sampling

2.1 Rejection sampling as a first example

A familiar example is the simplest form of rejection sampling. Draw $X_1, X_2, \dots$ independently from a full-support probability law π on a finite set Ω with $|\Omega| \geq 2$, reject a draw in $U \subseteq \Omega$, and return the first draw in $G = \Omega \setminus U$. This is the special case of the classical accept–reject method in which each proposed value is accepted or rejected deterministically (von Neumann, 1951; Devroye, 1986).

Write $p = \pi(U)$. If $0 < p < 1$, then

$$\mathbb{P}\{\tau \geq n\} = p^n, \qquad \mathbb{P}\{\tau = t\} = p^t(1-p), \qquad \mathbb{E}\tau = \frac{p}{1-p} \qquad (n, t \in \mathbb{Z}_{\geq 0}). \tag{3}$$

Here τ is the number of rejected draws; the returned value has law $\pi(\,\cdot \mid G)$. If $p = 0$, the first draw is accepted; if $p = 1$, the algorithm never stops. When $0 < p < 1$, the algorithm stops almost surely, although the set $U^{\mathbb{N}}$ of tapes on which it never stops is nonempty. Probability assigns this set mass zero. Effective dimension asks how much algorithmic randomness can nevertheless remain in one of its individual elements. The geometric survival law in (3) is standard; its role here is to provide the smallest case in which survival, individual-tape complexity, and set dimension can all be computed exactly.

First suppose that the source is uniform and $U \neq \varnothing$. Write $q = |\Omega| \geq 2$ and $M = |U|$, so $1 \leq M \leq q$. The probability that the first n blocks are rejected is

$$\left(\frac{M}{q}\right)^n,$$

and there are exactly M^n possible surviving prefixes. Each can be specified using $n \log_2 M + O(\log n)$ bits, whereas an unrestricted source prefix contains $n \log_2 q$ bits. The ratio is

$$\frac{\log_2 M}{\log_2 q}.$$

In the two-bit example

$$\Omega = \{00, 01, 10, 11\}, \qquad U = \{00, 01\}.$$

Every surviving block has its first bit fixed and its second bit free. At most half of the source information can remain, and suitable tapes attain this value.

For a computable full-support source $P = \pi^{\mathbb{N}}$, let $\omega_{1:n}$ denote the first n source symbols (or blocks). For a finite word w, let $[w]$ be the cylinder of tapes beginning with w, and write $P[w] = P([w])$. Define their self-information by

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}].$$

Let $K(w)$ be the length of the shortest program that prints w on a fixed universal prefix-free, or self-delimiting, machine (Li and Vitányi, 2019). More generally, $K(w \mid z)$ is the corresponding program length when z is supplied as auxiliary input. The effective strong P-dimension of a tape ω is

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) = \limsup_{n\to\infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \tag{4}$$

Replacing the limsup by a liminf defines the effective P-dimension $\dim_{\mathrm{eff},P}(\omega)$. For a fair binary source, write

$$\mathrm{Dim}_{\mathrm{eff}}(\omega) = \limsup_{n\to\infty} \frac{K(\omega_{1:n})}{n}.$$

This is the standard constructive strong dimension relative to a source measure (Athreya et al., 2007; Lutz, 2011); the definition itself is not new. The word *pointwise* here refers to one realized tape. A value near one means that the tape is nearly as incompressible as an ordinary draw from P. A smaller value means that continued execution has imposed structure on the tape.

For a set E of tapes, $\dim_H^P E$ denotes its Billingsley, or source-relative Hausdorff, dimension: in the Hausdorff covering construction, the cylinder $[w]$ is assigned diameter $P[w]$ (Billingsley, 1965; Lutz, 2011). Under a uniform q-ary source, this is ordinary Hausdorff dimension in the symbolic metric $d(\omega, \eta) = q^{-k}$, where k is the length of the longest common prefix of ω and η.

The three quantities answer different questions:

(i) $\mathbb{P}\{\tau \geq n\}$ is the probability of performing at least n rejection or repair steps;

(ii) $\mathrm{Dim}_{\mathrm{eff},P}(\omega)$ measures the randomness of one tape; and

(iii) $\dim_H^P \mathcal{N}$ measures the size of the set of all nonterminating tapes.

None of these notions is introduced here. The question is whether one fixed resampling computation links them and when all three can be computed exactly. The first is distributional, the second is pointwise, and the third is geometric. In general, effective strong dimension, which uses a limsup, is associated with packing dimension rather than Hausdorff dimension (Athreya et al., 2007). The equality between maximal effective strong dimension and Hausdorff dimension in the examples below follows from their comparison laws; it is not a general identity.

2.2 A comparison law

The following standard consequence of coding and the mass-distribution principle will be used for both finite and infinite state. For a closed set $F \subseteq \Omega^{\mathbb{N}}$, let

$$\mathcal{T}_n(F) = \{w \in \Omega^n : [w] \cap F \neq \varnothing\}$$

be its extendible prefixes of length n.

Proposition 2.1 (A comparison law on surviving tapes)**.** *Let $P = \pi^{\mathbb{N}}$ be a computable full-support source on a finite alphabet Ω with $|\Omega| \geq 2$, and let F be a nonempty closed set. Suppose that there are $\delta \in [0, 1]$, a computable probability law Q with $Q(F) = 1$, and numbers $c_n \geq 1$ satisfying $\log_2 c_n = o(n)$ such that*

$$c_n^{-1} P[w]^\delta \leq Q[w] \leq c_n P[w]^\delta \qquad \big(w \in \mathcal{T}_n(F)\big). \tag{5}$$

Then $\dim_H^P F = \delta$. Every $\omega \in F$ satisfies $\mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq \delta$, and every tape that is Martin–Löf random under Q satisfies

$$\dim_{\mathrm{eff},P}(\omega) = \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \delta.$$

Proof. Put $p_{\max} = \max_x \pi(x) < 1$. If $\theta > \delta$, then

$$\sum_{w \in \mathcal{T}_n(F)} P[w]^\theta \leq c_n p_{\max}^{n(\theta-\delta)} \sum_{w \in \mathcal{T}_n(F)} Q[w] = c_n p_{\max}^{n(\theta-\delta)} \longrightarrow 0.$$

The length-n cylinders therefore give $\dim_H^P F \leq \delta$. The lower bound is trivial if $\delta = 0$. Otherwise, fix $0 < s < \delta$. Then

$$Q[w] \leq c_n P[w]^\delta \leq c_n p_{\max}^{n(\delta-s)} P[w]^s.$$

The coefficient is bounded because $\log c_n = o(n)$. The mass-distribution principle gives $\dim_H^P F \geq s$, and letting $s \uparrow \delta$ proves the dimension identity.

Uniform coding under the computable law Q gives

$$K(w) \leq -\log_2 Q[w] + O(\log n) \leq \delta \log_2 \frac{1}{P[w]} + \log_2 c_n + O(\log n)$$

for every $w \in \mathcal{T}_n(F)$. Since $-\log_2 P[w] \geq n \log_2(1/p_{\max})$, division by source information proves the effective strong-dimension upper bound. Since F is closed and $Q(F) = 1$, every point outside F lies in a Q-null cylinder; hence every Q-Martin–Löf-random tape belongs to F. For such a tape, Levin–Schnorr and the other inequality in (5) give the matching lower bound. $\square$

Proposition 2.1 merely isolates the assumption used later; it is not a new dimension theorem. The real issue is whether one can construct a consistent law Q satisfying the two-sided comparison on every surviving prefix. Finite-state Perron–Frobenius theory supplies such a law under the assumptions in Appendix A; in infinite state it need not exist.

2.3 Exact rejection sampling

For rejection sampling, the tail in (i) is $\pi(U)^n$. When U is nonempty, the two dimension questions are governed by the number δ that solves

$$\sum_{x \in U} \pi(x)^\delta = 1. \tag{6}$$

Every tape rejected forever then satisfies

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq \delta \qquad (\omega \in U^{\mathbb{N}}). \tag{7}$$

To see that the bound is attained, write $\mathbf{1}\{\cdot\}$ for the indicator of the stated condition and set

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}.$$

For a computable law Q, a tape is Martin–Löf random if it avoids every effective Q-null test; equivalently, the Levin–Schnorr theorem gives $K(\omega_{1:n}) \geq -\log_2 Q[\omega_{1:n}] - O(1)$ for every n (Li and Vitányi, 2019). The displayed Q_δ is a probability law. A tape that is Martin–Löf random under $Q_\delta^{\mathbb{N}}$ attains equality. The corresponding Hausdorff calculation gives $\dim_H^P(U^{\mathbb{N}}) = \delta$. Under the uniform law, $\pi(U) = q^{\delta-1}$, so the tail exponent and the dimension determine one another. For a nonuniform law, they need not.

The following bounds show what remains true without uniformity. Let $p = \pi(U)$, let $\nu = \pi(\,\cdot \mid U)$ be the law of a rejected symbol, and write $\alpha = \log_2(1/p)$ for the survival exponent.

Proposition 2.2 (Entropy bounds for the rejection dimension)**.** *Assume* $0 < p < 1$*. If* $|U| = 1$*, then* $\delta = 0$*. If* $|U| \geq 2$*, define the Rényi entropy of order* $s \in (0,1)$ *by*

$$H_s(\nu) = \frac{1}{1-s} \log_2 \sum_{x \in U} \nu(x)^s,$$

and let $H(\nu)$ *be the Shannon entropy (Rényi, 1961). Then*

$$\delta = \frac{H_\delta(\nu)}{\alpha + H_\delta(\nu)} \tag{8}$$

and hence

$$\frac{H(\nu)}{\alpha + H(\nu)} \leq \delta \leq \frac{\log_2 |U|}{\alpha + \log_2 |U|}. \tag{9}$$

The two bounds coincide when ν is uniform on U.

Proof. For $|U| \geq 2$, the root lies in $(0, 1)$. Since $\pi(x) = p\nu(x)$ on U, the defining equation for δ becomes

$$1 = p^\delta \sum_{x \in U} \nu(x)^\delta.$$

Taking logarithms gives $\delta\alpha = (1 - \delta)H_\delta(\nu)$, which proves (8). The bounds follow from $H(\nu) \leq H_\delta(\nu) \leq \log_2 |U|$. □

Thus α is the information cost of surviving one more draw, while $H_\delta(\nu)$ measures the choice that remains among rejected symbols. Their balance determines the dimension. The law Q_δ produces tapes attaining the pointwise bound, and regularity of the product set makes the Hausdorff dimension equal to the same value.

3 Reversing a resampling computation

The probabilistic and coding step used below is elementary. It is stated separately so that the resampling-specific content of Theorem 3.2 is clear.

Proposition 3.1 (Likelihood and coding on a stopping line)**.** *Let $\mathcal{W}$ be a prefix-free set of source-tape prefixes, let $R : \mathcal{W} \to \mathcal{R}$ be injective, and let G be a subprobability mass on $\mathcal{R}$ that is positive on $R(\mathcal{W})$. Define*

$$\mu(w) = G(R(w)), \qquad \Delta(w) = \log_2 \frac{\mu(w)}{P_{\text{tape}}(w)}.$$

Then, for every $u \geq 0$,

$$P_{\text{tape}}\{w \in \mathcal{W} : \Delta(w) \geq u\} \leq 2^{-u}. \tag{10}$$

If μ is lower semicomputable uniformly from auxiliary data d, then

$$K(w \mid d) \leq -\log_2 P_{\text{tape}}(w) - \Delta(w) + O(1) \qquad (w \in \mathcal{W}). \tag{11}$$

Proof. Injectivity gives $\sum_{w \in \mathcal{W}} \mu(w) \leq \sum_r G(r) \leq 1$. The cylinders determined by $\mathcal{W}$ are disjoint, and hence

$$\sum_{\Delta(w) \geq u} P_{\text{tape}}(w) = \sum_{\Delta(w) \geq u} 2^{-\Delta(w)} \mu(w) \leq 2^{-u}.$$

The coding theorem for lower semicomputable subprobability masses gives $K(w \mid d) \leq -\log_2 \mu(w) + O(1)$, which is (11). □

The proposition is a change-of-measure and Kraft-coding fact. The work in a resampling application is to construct an injective backward record for which the likelihood ratio has a useful expression.

3.1 The resampling factorization

Let $X_1, \dots, X_n$ be independent finite-valued variables with product law $\pi = \bigotimes_i \pi_i$. For $J \subseteq \{1, \dots, n\}$, write $\pi_J = \bigotimes_{i \in J} \pi_i$ for the marginal product law on the coordinates in J. Each nonempty bad event A is determined by the coordinates $v(A)$ and has probability $p_A > 0$. Its *marginal surprisal* is $\log_2(1/p_A)$. Fix a deterministic rule for selecting a currently true bad event.

List the random choices in the order in which the algorithm uses them. First draw $S_0 \sim \pi$. Whenever the current assignment contains a bad event, select A_t, draw $Z_t \sim \pi_{v(A_t)}$ independently of the past, and write Z_t on $v(A_t)$. Let S_t denote the resulting assignment after the tth repair. Let E_T be the event that at least T resamplings occur, and put

$$\Xi_T = (S_0, Z_1, \dots, Z_T).$$

Let P_{tape} denote the law generated by the initial draw and these successive independent resampling draws. Then $\mathbb{P}(E) = P_{\text{tape}}(E)$ for every tape event E; for a finite used-tape prefix w (including Ξ_T), $P_{\text{tape}}(w)$ denotes the probability of its cylinder. Thus $E_T = \{\tau \geq T\}$. For each fixed T, the minimal tape prefixes that realize T resamplings form a prefix-free family: no member is a proper prefix of another. Their cylinders are therefore disjoint, and their probabilities sum to $\mathbb{P}(E_T)$.

Let $Y_t = S_{t-1}|_{v(A_t)}$ be the configuration overwritten at time t, and let $L_T = (A_1, \dots, A_T)$. Write $\pi_{v(A)}(\,\cdot \mid A)$ for the marginal law $\pi_{v(A)}$ conditioned on the local configurations satisfying A. Coordinatewise, the entries of the used tape are partitioned into the final assignment and the overwritten configurations. Therefore

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^{T} p_{A_t}\, \pi_{v(A_t)}(Y_t \mid A_t). \tag{12}$$

Let Q_T be a subprobability law on repair histories $\ell = (A_1, \dots, A_T)$, positive on every feasible history of length T. Thus $-\log_2 Q_T(\ell)$ is its ideal code length. Define the backward subprobability law on complete records by

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^{T} \pi_{v(A_t)}(y_t \mid A_t). \tag{13}$$

The map

$$\Xi_T \longmapsto (S_T, L_T, Y_{1:T})$$

is injective. Indeed, reverse the repairs from T to 1: the current values on $v(A_t)$ recover Z_t, and restoring Y_t recovers the preceding assignment.

Theorem 3.2 (Backward likelihood identity)**.** *Every execution in E_T satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \tag{14}$$

Call the right side Δ_T on E_T and set $\Delta_T = -\infty$ otherwise. Then, for every $u \geq 0$,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \tag{15}$$

If the instance and the family $(Q_T)_{T \geq 1}$ are uniformly computable, then

$$-\log_2 P_{\text{tape}}(\Xi_T) - K(\Xi_T \mid T, \text{instance}) \geq \Delta_T - O(1). \tag{16}$$

Here instance *denotes a fixed effective description of the source laws, bad events, and deterministic selection rule. Uniform computability means that one algorithm can approximate $Q_T(\ell)$ to any requested precision from T and ℓ, with the instance data encoded effectively in the same sense.*

Proof. Dividing (13) by (12) cancels the final-state factor and all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\mathrm{tape}}(\Xi_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^{T} p_{A_t}} = 2^{\Delta_T},$$

which proves (14).

Let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes that produce T repairs. For $w \in \mathcal{W}_T$, put

$$R_T(w) = \big(S_T(w), L_T(w), Y_{1:T}(w)\big).$$

Backward reconstruction makes R_T injective. Moreover,

$$\sum_r G_T(r) = \sum_\ell Q_T(\ell) \le 1.$$

Proposition 3.1, with $G = G_T$ and $R = R_T$, now proves (15). For a finite effective instance, the stopping line and record map are uniformly computable from T; uniform computability of Q_T therefore makes the pulled-back mass $\mu_T(w) = G_T(R_T(w))$ uniformly lower semicomputable. Applying (11) with auxiliary data $(T, \mathrm{instance})$ proves (16) (Shannon, 1948; Fano, 1949; Li and Vitányi, 2019). □

Corollary 3.3 (The exact gap to the conditional survival law)**.** *Assume $\mathbb{P}(E_T) > 0$. Let $\mathcal{W}_T$ be the stopping line in the proof of Theorem 3.2, put*

$$P_T(w) = \frac{P_{\mathrm{tape}}(w)}{\mathbb{P}(E_T)}, \qquad \mu_T(w) = G_T(R_T(w)), \qquad m_T = \sum_{w \in \mathcal{W}_T} \mu_T(w),$$

so that $0 < m_T \le 1$, and let $\bar\mu_T(w) = \mu_T(w)/m_T$. Then

$$\log_2 \frac{1}{\mathbb{P}(E_T)} = \mathbb{E}_{P_T}[\Delta_T] + D_2(P_T \| \bar\mu_T) + \log_2 \frac{1}{m_T}, \tag{17}$$

where D_2 is relative entropy with base-two logarithms. In particular,

$$\mathbb{E}[\Delta_T \mid E_T] \le \log_2 \frac{1}{\mathbb{P}(E_T)}. \tag{18}$$

Equality holds precisely when $m_T = 1$ and the pulled-back backward law equals the tape law conditioned on E_T.

Proof. Since $\Delta_T(w) = \log_2\{\mu_T(w)/P_{\mathrm{tape}}(w)\}$,

$$D_2(P_T \| \bar\mu_T) = -\mathbb{E}_{P_T}[\Delta_T] + \log_2 \frac{m_T}{\mathbb{P}(E_T)}.$$

Rearranging proves (17). The inequality and its equality condition follow from $D_2(P_T \| \bar\mu_T) \ge 0$ and $m_T \le 1$. □

Theorem 3.2 immediately gives the normalized bound

$$\mathbb{P}\left\{\frac{\Delta_T}{T} \ge a\right\} \le 2^{-aT} \qquad (T \ge 1,\ a \ge 0). \tag{19}$$

The likelihood identity also gives an exact criterion for the universal upper bound to have a matching exponential lower bound. Let $\mu_T(w) = G_T(R_T(w))$ and, for $a, \varepsilon \geq 0$, define

$$A_T(a, \varepsilon) = \left\{ w \in \mathcal{W}_T : a \leq \frac{\Delta_T(w)}{T} \leq a + \varepsilon \right\}.$$

Then $P_{\text{tape}}(w) = 2^{-\Delta_T(w)} \mu_T(w)$ gives

$$2^{-(a+\varepsilon)T} \mu_T\big(A_T(a, \varepsilon)\big) \leq P_{\text{tape}}\big(A_T(a, \varepsilon)\big) \leq 2^{-aT} \mu_T\big(A_T(a, \varepsilon)\big). \tag{20}$$

Consequently,

$$-\frac{1}{T} \log_2 \mathbb{P}\left\{ \frac{\Delta_T}{T} \geq a \right\} \longrightarrow a \tag{21}$$

if and only if

$$-\frac{1}{T} \log_2 \mu_T\big(A_T(a, \varepsilon)\big) \longrightarrow 0 \qquad \text{for every fixed } \varepsilon > 0.$$

Here $-\log_2 0 = +\infty$. If the displayed condition holds, (20) places the limiting tail rate between a and $a + \varepsilon$; let $\varepsilon \downarrow 0$. Conversely, if the tail at a is $2^{-aT+o(T)}$, the universal bound above $a + \varepsilon$ shows that the band itself has probability $2^{-aT+o(T)}$. Its upper bound in (20) then forces the band to have subexponential μ_T-mass. There is no unconditional lower bound because the relevant backward mass can be exponentially small or zero.

The constant and exponent in (19) cannot be improved. For any $a > 0$, a one-event rejection sampler with event probability $p = 2^{-a}$ and $Q_T = 1$, one has $\Delta_T = aT$ on E_T, and hence

$$\mathbb{P}\left\{ \frac{\Delta_T}{T} \geq a \right\} = \mathbb{P}(E_T) = p^T = 2^{-aT}. \tag{22}$$

Large-deviation interpretation. No large-deviation principle is assumed in the paper. If Δ_T/T independently satisfies one with a base-two rate function J, and the principle identifies the exact upper-tail rate at a, then

$$\inf_{x \geq a} J(x) \geq a. \tag{23}$$

The one-event example shows that equality is possible; subexponential backward mass in the band gives equality more generally. For ordinary rejection sampling with rejection probability p, survival means that T independent Bernoulli variables all equal one, so its rate is the endpoint relative entropy

$$D_2(1 \| p) = \log_2 \frac{1}{p}.$$

For the finite-state model in Appendix A, the corresponding escape rate is $-\log_2 \rho(B_1)$ (Dembo and Zeitouni, 2010; Keller and Liverani, 2009).

The two terms in (14) have different origins. The first sums the marginal surprisals of the events that were repaired; the second is the ideal code length of their ordered history. The first sum may count statistically dependent information more than once. Neither term alone gives a bound. Their difference does, because it is a likelihood ratio.

For example, color the vertices of a graph independently with q colors and repeatedly recolor the endpoints of the first monochromatic edge. A fixed edge is monochromatic with probability $1/q$, so each repair contributes $\log_2 q$ bits to the first term. The second term pays for the repaired edges and their order. The comparison is therefore between information ruled out by the violations and information used by the algorithm's history.

Corollary 3.4 (A tail bound and a pointwise dimension bound)**.** *Suppose every T-repair execution satisfies*

$$\sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} \geq \beta T - C_0, \qquad -\log_2 Q_T(L_T) \leq \gamma T + C_1,$$

where $\beta > \gamma$. Then

$$\mathbb{P}(E_T) \leq 2^{C_0 + C_1 - (\beta-\gamma)T}. \tag{24}$$

For the pointwise conclusion, assume in addition that

(i) the instance and the family $(Q_T)_{T\geq 1}$ are uniformly computable;

(ii) the computation reads a fixed fair binary tape sequentially, the initial draw consumes $c_0 = O(1)$ bits, and every repair consumes exactly c bits, so that Ξ_T is uniformly computably interconvertible with $\omega_{1:c_0+cT}$.

Then every nonterminating tape satisfies

$$\mathrm{Dim}_{\mathrm{eff}}(\omega) \leq \max\left\{0, 1 - \frac{\beta - \gamma}{c}\right\}. \tag{25}$$

Proof. The assumptions give $\Delta_T \geq (\beta - \gamma)T - (C_0 + C_1)$. Equation (24) follows from (15). Equation (16), together with the $O(\log_2 T)$ bits needed to specify T, gives $K(\Xi_T) \leq c_0 + cT - (\beta - \gamma)T + o(T)$ along an infinite execution. If $\beta - \gamma > c$, this is impossible for all large T, so no infinite run exists. Otherwise divide by $c_0 + cT$; the gap between consecutive repair prefixes is the fixed number c of bits. □

Remark 3.5 (What the record must contain). For a recursive Moser-style implementation, L_T must be enlarged to include the recursion tree or return structure unless those data are recoverable by a separate prefix-free decoding rule. In Theorem 3.2, everything needed for reversal contributes to the probability $Q_T(L_T)$.

4 Beyond finite state

Proposition 3.1 itself does not require a finite state space: it applies whenever the live prefixes form an effective stopping line and the backward record is injective. Finite state is one general condition under which a comparison law on infinite surviving tapes can be obtained from a transition matrix; the next example shows that such a law may also exist with countably many states.

Finiteness is not merely a convenience in the pressure calculation of Appendix A. If a finite substochastic Markov chain is absorbed almost surely, then its survival probability decays exponentially. With countably many states, probability can drift toward states from which absorption takes progressively longer. Almost-sure termination may then coexist with a polynomial tail and a nonterminating set of full dimension.

4.1 A null set of full dimension

Let $\xi_1, \xi_2, \ldots$ be independent and uniform on $\{-1, +1\}$. Starting from $X_0 = 1$, set

$$X_n = 1 + \sum_{i=1}^{n} \xi_i, \qquad \tau = \inf\{n \geq 1 : X_n = 0\},$$

and halt when the walk reaches zero. Write $E_n = \{\tau > n\}$ and $\mathcal{N} = \{\tau = \infty\}$.

Proposition 4.1 (Polynomial survival at full dimension). *The killed walk satisfies*

$$\mathbb{P}(E_n) = 2^{-n}\binom{n}{\lfloor n/2 \rfloor} \sim \sqrt{\frac{2}{\pi n}}. \tag{26}$$

It therefore halts almost surely and has survival exponent zero. Nevertheless,

$$\dim_H \mathcal{N} = 1, \qquad \max_{\omega \in \mathcal{N}} \dim_{\mathrm{eff},P}(\omega) = \max_{\omega \in \mathcal{N}} \mathrm{Dim}_{\mathrm{eff},P}(\omega) = 1, \tag{27}$$

where P is the fair binary source.

Proof. Writing $S_k = X_k - 1$, the reflection principle gives, for every reachable $r \geq 0$ of the same parity as n,

$$\#\{w : S_k(w) \geq 0 \ (1 \leq k \leq n),\ S_n(w) = r\} = \binom{n}{(n+r)/2} - \binom{n}{(n+r)/2+1}.$$

Summing over r telescopes to $\binom{n}{\lfloor n/2 \rfloor}$. Equation (26) follows from Stirling's formula (Feller, 1968).

To determine the exceptional set, use the positive harmonic function $h(x) = x$ for the walk killed at zero. Its Doob transform is

$$Q(x, x+1) = \frac{x+1}{2x}, \qquad Q(x, x-1) = \frac{x-1}{2x}. \tag{28}$$

At $x = 1$ the second probability is zero. These transitions define a computable law supported on $\mathcal{N}$, and telescoping gives, for every live prefix w of length n,

$$Q[w] = 2^{-n} X_n(w) = P[w] X_n(w). \tag{29}$$

Since $1 \leq X_n(w) \leq n+1$, condition (5) holds with $\delta = 1$ and $c_n = n+1$. Proposition 2.1 proves (27). Equivalently, every Q-Martin–Löf-random tape satisfies

$$K(\omega_{1:n}) \geq n - \log_2(n+1) - O(1),$$

and hence has effective and effective strong dimension one. The transform in (28) is the classical Doob law usually described as $S_n = X_n - 1$ conditioned to stay nonnegative, equivalently as X conditioned in the Doob-transform sense never to hit zero (Bertoin and Doney, 1994). □

This example separates exponential scale from finite-horizon information. Although $\alpha = 0$ and $\dim_H \mathcal{N} = 1$,

$$-\log_2 \mathbb{P}(E_n) = \frac{1}{2}\log_2 n + O(1). \tag{30}$$

Conditioned on E_n, the live prefixes are uniform. Their conditional law $Q_n^\star$ therefore satisfies, for every live w,

$$\log_2 \frac{Q_n^\star[w]}{P[w]} = -\log_2 \mathbb{P}(E_n), \qquad K(w \mid n) \leq n - \frac{1}{2}\log_2 n + O(1). \tag{31}$$

This coding bound is sharp at the logarithmic scale. Put $N_n = \binom{n}{\lfloor n/2 \rfloor}$. Since fewer than 2^k strings have conditional prefix complexity below k, for every $r \geq 1$,

$$Q_n^\star\{w : K(w \mid n) < \log_2 N_n - r\} \leq 2^{-r}. \tag{32}$$

Therefore, for all but a 2^{-r} fraction of the surviving prefixes, $K(w \mid n) = n - \frac{1}{2}\log_2 n + O(r+1)$. This statement holds separately at each horizon; it does not assert that one infinite tape attains the estimate at every n.

Thus the likelihood comparison retains the logarithmic cost that ordinary dimension discards. The horizon-conditioned laws $Q_n^\star$ are not consistent as n varies; the Doob law Q is consistent, but its likelihood ratio depends on the endpoint through X_n. The endpoint factors in (29) and (52) reveal the common mechanism: the finite-state factor is bounded, while the random-walk factor grows only polynomially. Both disappear after normalization by n, although the latter carries the logarithmic correction in (30).

Although $\mathcal{N}$ has dimension one, it contains no tape that is Martin–Löf random under the fair source. Indeed, the sets E_n are uniformly computable and their probabilities decrease effectively to zero, so a subsequence of them forms a Martin–Löf test. Effective dimension one is weaker than full Martin–Löf randomness.

4.2 The unrestricted boundary

A *computable prefix-reading machine* is a deterministic Turing machine with a one-way read-only binary input tape. Its work tape gives it countably many effectively represented configurations, and it may perform finite internal computation between successive input reads.

Proposition 4.2 (Nontermination and effectively closed sets)**.** *The nontermination sets of computable prefix-reading machines are exactly the Π^0_1, or effectively closed, subsets of Cantor space.*

Proof. If a machine halts, its finite computation inspects only a finite tape prefix. The set of tapes on which it halts is therefore effectively open, so its nontermination set is effectively closed.

Conversely, write an effectively closed set as $F = [T]$, where T is a computable prefix-closed binary tree. After reading each new bit, a machine tests whether the prefix read so far belongs to T and halts when it does not. It runs forever exactly on the infinite paths through T, namely on F. $\square$

This is the standard representation of effectively closed classes by computable trees (Downey and Hirschfeldt, 2010). It shows that countable state alone imposes essentially no geometric regularity: even simple computably presented closed subsets of Cantor space can have noncomputable Hausdorff dimension (Staiger, 2020). Consequently, no algorithm can compute the exact nontermination dimension from an arbitrary prefix-reading machine description.

The useful problem lies between finite automata and unrestricted machines. The finite-state Perron law and the random-walk Doob transform both produce one measure satisfying (5). For countable-state resampling, the substantive question is when the finite-horizon laws produced by backward reconstruction can be replaced by a consistent computable law without losing their likelihood bounds. Merely replacing a finite matrix by an infinite one does not settle this question. Classical Perron theory for countable nonnegative matrices and thermodynamic formalism for countable Markov shifts already develop the corresponding transient, null-recurrent, and positive-recurrent classifications; outside the positive-recurrent regime, the finite-state Perron construction need not produce a normalizable equilibrium law (Vere-Jones, 1967; Sarig, 1999).

5 What a dependency graph forgets

To use Theorem 3.2, one must describe the possible lists of repairs. The Lovász local lemma summarizes event interactions by a dependency graph, and many algorithmic proofs use that graph to organize repair histories. This section asks what those graph data determine about the exact tail and the nonterminating tapes of a fixed computation.

5.1 Why dependency graphs enter

Let $\mathcal{A}$ be a finite family of bad events. The Lovász local lemma asks whether all of them can be avoided. In the variable model, where independent variables generate the events, the standard dependency graph D joins two events whenever their variable sets overlap. For a fixed graph, Shearer's region is the sharp worst-case avoidability criterion based only on D and the marginal event probabilities (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005).

The Moser–Tardos algorithm repeatedly selects a true bad event and redraws the variables on which it depends. Its analyses encode possible repair histories as trees, directed acyclic graphs, or entropy-compression records whose weights are determined by event probabilities and adjacency in D (Moser and Tardos, 2010; Haeupler and Harris, 2017; Fialho et al., 2022).

The graph records which repairs may interact, but not $\mathbb{P}(A \cap B)$, set inclusion, or $\mathbb{P}(\bigcup_A A)$. Even if all pairwise intersection probabilities are supplied separately, higher-order intersections can still change the union. The remainder of the section compares the exact state-space calculation with graph-based counting and then gives two systems for which this missing higher-order information changes the answer.

5.2 The full state space

Every finite variable-model resampling algorithm is a finite-state randomized computation on assignments. Set $P = P_{\text{tape}}$, the sequential source law of the initial assignment and subsequent resampling draws; finite initial factors do not affect the dimensions below. Delete zero-probability outcomes and assume that every surviving source outcome has probability strictly less than one. Fix a deterministic rule for selecting a currently true bad event. On reachable nonterminal assignments and for $\theta \geq 0$, define

$$(B_\theta)_{xy} = \sum_{z\colon F(x,z)=y} \pi_{v(A(x))}(z)^\theta, \tag{33}$$

where $A(x)$ is the selected event and $F(x,z)$ is the updated assignment. Decompose the nonterminal transition graph into reachable strongly connected components C containing a directed cycle, where a strongly connected component is a maximal set of states that are mutually reachable by directed paths. Let $B_{\theta,C}$ be the restriction of B_θ to C. If no such component exists, then $\mathcal{N} = \varnothing$, its dimension is zero, and $\alpha = +\infty$ because survival becomes impossible after finitely many steps. Otherwise, Proposition A.1 gives

$$\alpha = -\log_2 \max_C \rho(B_{1,C}), \qquad \dim_H^P \mathcal{N} = \max_C \delta_C, \qquad \rho(B_{\delta_C,C}) = 1. \tag{34}$$

Here $\delta_C \in [0,1]$ is the unique solution of the displayed spectral-radius equation for component C. Under computability,

$$\sup_{\omega\in\mathcal{N}} \operatorname{Dim}_{\text{eff},P}(\omega) = \dim_H^P \mathcal{N} = \max_C \delta_C. \tag{35}$$

Every infinite path eventually remains in one component; the possible finite entry paths form a countable union and change neither the exponential rate nor either dimension. Thus the full assignment gives all three quantities exactly, but its matrix usually has exponentially many rows and columns.

5.3 The graph bound

Classical entropy-compression and Shearer arguments use the dependency graph and the probabilities p_A to bound the total weight of possible repair lists (Cartier and Foata, 1969; Scott and Sokal, 2005). Only the complete-graph case is needed here. If $D = K_m$ is complete on $m = |\mathcal{A}|$ event vertices, the graph-only enumeration allows every word $(A_1, \ldots, A_t) \in \mathcal{A}^t$ and assigns it weight $\prod_{i=1}^t p_{A_i}$. Its total weight is

$$\left(\sum_{A \in \mathcal{A}} p_A \right)^t.$$

Put $S = \sum_A p_A$ and choose the history law

$$Q_T(A_1, \ldots, A_T) = \frac{\prod_{t=1}^T p_{A_t}}{S^T}$$

on all event words. Theorem 3.2 then gives $\Delta_T = -T \log_2 S$. If $S < 1$, it follows that

$$\mathbb{P}(E_T) \leq S^T. \tag{36}$$

For a computable uniform whole-block system on $\Omega = [q]^b$, in which every repair redraws all b coordinates, let P be the product source on blocks. Summing the compatible overwritten blocks over all event words gives at most

$$\sum_{(A_1,\ldots,A_T)} \prod_{t=1}^T |A_t| = \left(q^b S\right)^T$$

length-T overwritten-block strings. The final block contributes only the fixed factor q^b, which disappears after normalization. Coding the resulting prefixes gives

$$d_{\text{graph}} = \left[1 + \frac{\log_2 S}{b \log_2 q}\right]_{[0,1]}, \qquad \operatorname{Dim}_{\text{eff},P}(\omega) \leq d_{\text{graph}} \quad (\omega \in \mathcal{N}), \quad \dim_H^P \mathcal{N} \leq d_{\text{graph}}, \tag{37}$$

where $[\cdot]_{[0,1]}$ denotes truncation to $[0, 1]$. These are graph-based upper bounds; they need not equal the survival probability or dimension for a fixed selection rule. A short graph description also does not make exact computation easy: the relevant independence polynomial is #P-hard to evaluate in general (Harvey et al., 2018).

5.4 Whole-block resampling

Let Ω be a finite block space with source law π satisfying $0 < \pi(x) < 1$ for every x, and let $\mathcal{A}$ be any finite family of bad subsets. Every repair redraws the whole block from π. Fix a deterministic selector ϕ that, for each $x \in U = \bigcup_{A \in \mathcal{A}} A$, chooses one event $\phi(x)$ containing x. The selector partitions the bad union into the disjoint regions

$$C_A = \{x \in U : \phi(x) = A\}. \tag{38}$$

Let $\mathcal{I} = \{A \in \mathcal{A} : C_A \neq \varnothing\}$ index the nonempty regions, and define

$$W(\theta) = \sum_{x \in U} \pi(x)^\theta. \tag{39}$$

Theorem 5.1 (Exact whole-block calculation). *Assume $U \neq \varnothing$, and let $P = \pi^{\mathbb{N}}$. There is a unique $\delta \in [0,1]$ such that*

$$W(\delta) = \sum_{x \in U} \pi(x)^{\delta} = 1. \tag{40}$$

If the alphabet, π, and U are computable, then every nonterminating tape satisfies

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq \delta \qquad (\omega \in \mathcal{N} = U^{\mathbb{N}}). \tag{41}$$

The law

$$Q_{\delta}(x) = \pi(x)^{\delta} \mathbf{1}\{x \in U\} \tag{42}$$

is a probability law, and every tape that is Martin–Löf random for $Q_{\delta}^{\mathbb{N}}$ satisfies $\dim_{\mathrm{eff},P}(\omega) = \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \delta$. Moreover,

$$\dim_H^P \mathcal{N} = \delta, \qquad \mathbb{P}\{\tau \geq t\} = \pi(U)^t. \tag{43}$$

Under the uniform law on $\Omega = [q]^b$, writing $M = |U|$,

$$\dim_H \mathcal{N} = \delta = \frac{\log_2 M}{b \log_2 q}, \qquad \mathbb{P}\{\tau \geq t\} = \left(\frac{M}{q^b}\right)^t. \tag{44}$$

Proof. The function W is continuous and strictly decreasing because $0 < \pi(x) < 1$. Moreover, $W(0) = |U| \geq 1$ and $W(1) = \pi(U) \leq 1$, so (40) has a unique solution in $[0,1]$.

Every repair is a fresh draw from π and the computation continues exactly when that draw lies in U. Hence $\mathcal{N} = U^{\mathbb{N}}$ and $\mathbb{P}\{\tau \geq t\} = \pi(U)^t$. The law in (42) satisfies $Q_{\delta}^{\mathbb{N}}[w] = P[w]^{\delta}$ for every $w \in U^n$. Proposition 2.1, with $c_n = 1$, proves the pointwise and Hausdorff statements. The uniform formulas follow by substituting $\pi(x) = q^{-b}$. □

Theorem 3.2 recovers the survival law without a state matrix. Relabel each repair by the disjoint region C_A containing the overwritten block, put $r_A = \pi(C_A)$, and choose

$$Q_T(A_1, \ldots, A_T) = \prod_{t=1}^{T} \frac{r_{A_t}}{\pi(U)}.$$

For the relabeled event C_{A_t}, the first term in (14) is $\log_2(1/r_{A_t})$, whereas encoding the region label under Q_T costs $\log_2(\pi(U)/r_{A_t})$ bits. Their difference is $\log_2(1/\pi(U))$ at every repair. Conditioning only on the original event A_t can count overlapping mass more than once.

Comparison with the dependency graph. Under the standard variable-overlap definition, all common-block events are pairwise adjacent. The graph calculation therefore uses the one-step weight $\sum_{A \in \mathcal{A}} \pi(A)$, while the true rejection probability is $\pi(U)$. Under the uniform source on $[q]^b$, the resulting upper bound and the true dimension are

$$1 + \frac{\log_2 \sum_{A \in \mathcal{A}} \pi(A)}{b \log_2 q} \quad \text{and} \quad 1 + \frac{\log_2 \pi(U)}{b \log_2 q}, \tag{45}$$

again truncated to $[0,1]$. Before truncation, the two expressions agree when the events are disjoint, whereas overlap produces overcounting. Truncation can conceal this difference when both values are at least one.

In the variable model, a nonedge certifies disjoint scopes, whereas an edge marks only possible interaction. Nested Hamming balls give a simple overcount. Under the uniform law on $\{0,1\}^b$, let

$U_r = \{x : |x| \le r\}$ and $M_r = |U_r|$. The true dimension is $b^{-1} \log_2 M_r$. If the stricter rule $U_s \subset U_r$ is also recorded, the graph sum uses $(M_s + M_r)/2^b$ although the rejected set still has mass $M_r/2^b$. Taking $r = b - 1$ and $s = b - 2$ makes the graph bound trivial while the true dimension remains $b^{-1} \log_2(2^b - 1) < 1$.

Proposition 5.2 (Pairwise event data do not determine nontermination)**.** *There are two whole-block resampling systems with the same event–variable incidence graph, dependency graph, ordered event probabilities, and all pairwise intersection probabilities, but different survival exponents, maximal effective strong dimensions, and Hausdorff dimensions of nontermination.*

Proof. Let π be uniform on $\Omega = \{0,1\}^4$, and define

$$\begin{aligned} A &= \{0000, 0010, 0100\}, \quad & B &= \{0000, 0110, 1000\}, \\ C &= \{0010, 0110, 1010\}, \quad & C' &= \{0000, 1010, 1100\}. \end{aligned}$$

Membership in each event depends essentially on all four coordinates. Consequently, the systems $\{A, B, C\}$ and $\{A, B, C'\}$ have the same event–variable incidence graph $K_{3,4}$ and dependency graph K_3. Every event has probability $3/16$, and the two systems have the same pairwise intersection data:

$$\pi(A \cap B) = \pi(A \cap C) = \pi(B \cap C) = \pi(A \cap C') = \pi(B \cap C') = \frac{1}{16}.$$

Their triple intersections differ:

$$A \cap B \cap C = \varnothing, \qquad A \cap B \cap C' = \{0000\}.$$

Inclusion–exclusion therefore gives

$$|A \cup B \cup C| = 6, \qquad |A \cup B \cup C'| = 7.$$

In each system, select the first true event in the displayed order and redraw the entire block. Since every repair redraws the same four coordinates, the choice on an intersection does not affect survival. The two survival probabilities are

$$\left(\frac{3}{8}\right)^t \quad \text{and} \quad \left(\frac{7}{16}\right)^t,$$

with survival exponents

$$\alpha_1 = \log_2 \frac{8}{3}, \qquad \alpha_2 = \log_2 \frac{16}{7}.$$

Let $\mathcal{N}_1 = (A \cup B \cup C)^{\mathbb{N}}$ and $\mathcal{N}_2 = (A \cup B \cup C')^{\mathbb{N}}$. Theorem 5.1 gives

$$\max_{\omega \in \mathcal{N}_1} \mathrm{Dim}_{\mathrm{eff}}(\omega) = \dim_H \mathcal{N}_1 = \frac{\log_2 6}{4}, \qquad \max_{\omega \in \mathcal{N}_2} \mathrm{Dim}_{\mathrm{eff}}(\omega) = \dim_H \mathcal{N}_2 = \frac{\log_2 7}{4}.$$

Thus even the incidence graph, marginals, and all pairwise intersection probabilities fail to determine the exact survival exponent, maximal effective strong dimension, or Hausdorff dimension. $\square$

The set-theoretic mechanism is elementary inclusion–exclusion: pairwise data do not fix a triple intersection. Its consequence here is that the same omission separates the exact survival law and both notions of tape dimension at once.

Relation to the variable local lemma. The separation does not contradict the tightness of Shearer's region, which is a worst-case avoidability statement. The common marginal vector above satisfies $\pi(A) + \pi(B) + \pi(C) = 9/16 < 1$ (with C' replacing C in the second system), so both systems lie strictly inside the region for K_3. Variable-model and intersection-sensitive refinements give stronger avoidability or convergence guarantees (He et al., 2017, 2023). The proposition above has a different output: it concerns the exact survival law and the exceptional tapes of a specified process. It is neither a consequence nor a strengthening of the variable local lemma.

5.5 Partially overlapping local repairs

The preceding examples are memoryless because every action redraws the same block. A partially overlapping local repair leaves values relevant to later repairs unchanged, and those retained values can affect which event is selected next.

Existing representations of repair histories retain some of this persistent state information (Achlioptas and Iliopoulos, 2016; Haeupler and Harris, 2017; Fialho et al., 2022). The full assignment gives the exact formulas (34)–(35), but has exponentially many states. A repair label is smaller, but need not contain enough information to determine the next repair or reconstruct the preceding assignment.

Open problem. For a q-ary resampling instance of size n whose event–variable incidence graph has pathwidth w, find conditions under which a computable state space of size $q^{O(w)}\,\mathrm{poly}(n)$ both determines the next repair and permits reversal of every run, and whose weighted transition matrix gives the exact survival exponent and nontermination dimension. A spectral calculation alone does not give the pointwise bound, whereas a description that represents one run more than once need not give the exact Hausdorff dimension.

6 Conclusion

The survival tail measures how rapidly prolonged runs become rare; effective dimension measures the randomness of one nonterminating tape; and Hausdorff dimension measures the size of the entire nonterminating set. The backward law directly connects the first two. Its log-likelihood ratio is exactly the marginal information in the repaired events minus the information needed to describe their order. Bounding that ratio gives a running-time tail, while coding the same law bounds the Kolmogorov complexity of each surviving tape prefix. When the finite-horizon laws extend to one comparison law on infinite surviving tapes, the same method also determines Hausdorff dimension.

Rejection sampling gives the sharp exponential case. The killed random walk gives a complementary infinite-state case: it terminates almost surely with a polynomial tail, although its nonterminating set is null and has full dimension. The logarithmic finite-horizon cost of survival is invisible to ordinary dimension but remains visible in the likelihood ratio. Separately, the matched whole-block systems show that event probabilities and dependency structure even when supplemented by all pairwise intersection probabilities do not determine the exact survival law, the maximal effective strong dimension, or the Hausdorff dimension of a fixed resampling process.

Because unrestricted machines realize every effectively closed set, exact dimension cannot in general be computed from a machine description. The remaining task is to identify structure that makes the exact calculation tractable. For local resampling, this means finding a state small enough to compute with but rich enough to determine the next repair and reconstruct the preceding one. In countable state, it means determining when there is one computable law on surviving paths whose likelihood ratio gives matching probability, pointwise-complexity, and Hausdorff-dimension bounds.

A Finite-state computations

This appendix records the classical finite-state pressure calculation used in the text. Its role is to fix notation and to identify precisely what finite state supplies; the calculation itself is not a new contribution.

Let $\mathcal{G}$ be a finite directed multigraph of nonterminal states with fixed initial state s_0. Each outgoing edge $e : s \to t$ represents a distinct next source outcome and has probability $p(e) > 0$. The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path $w = e_1 \cdots e_n$, traversing states $s_0, s_1, \ldots, s_n$, determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^{n} p(e_i).$$

Let $\mathcal{W}_n$ be the set of surviving paths of length n from s_0. Assume that the reachable nonterminal graph is strongly connected, contains a directed cycle, and satisfies $\max_e p(e) < 1$. For $\theta \geq 0$, define

$$(B_\theta)_{st} = \sum_{e:s\to t} p(e)^\theta, \qquad \Lambda(\theta) = \log_2 \rho(B_\theta). \tag{46}$$

For any square matrix C, its spectral radius is $\rho(C) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } C\}$. At $\theta = 1$, B_1 is the substochastic transition matrix among nonterminal states: each row sum is at most one, and the missing mass is the one-step termination probability. The quantity $\Lambda(\theta)$ is the base-two pressure, the exponential growth rate of the total θ-weight of surviving paths. Graphs that are not strongly connected are handled by applying the result to each reachable strongly connected component C containing a directed cycle. If $\rho(B_{\delta_C,C}) = 1$, then

$$\alpha = -\log_2 \max_C \rho(B_{1,C}), \qquad \dim_H^P \mathcal{N} = \max_C \delta_C. \tag{47}$$

Under computability, the last quantity is also $\sup_{\omega\in\mathcal{N}} \mathrm{Dim}_{\mathrm{eff},P}(\omega)$. Every infinite path eventually remains in one component; the possible finite entry paths form a countable union and change neither the exponential rate nor either dimension. If there is no reachable cyclic component, the nonterminating set is empty and $\alpha = +\infty$.

Use $P[w]$ as the diameter of the cylinder $[w]$. The resulting Hausdorff dimension is the source-relative dimension $\dim_H^P$ defined in the introduction; for a uniform q-ary source it is ordinary Hausdorff dimension in the q-adic metric. For an infinite path $\omega = e_1 e_2 \cdots$, use the corresponding pointwise definition

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) = \limsup_{n\to\infty} \frac{K(e_1 \cdots e_n)}{-\log_2 P[e_1 \cdots e_n]},$$

and define $\dim_{\mathrm{eff},P}$ by replacing the limsup with a liminf.

Proposition A.1 (The finite-state formula)**.** *There is a unique $\delta \in [0,1]$ satisfying $\rho(B_\delta) = 1$. Moreover,*

$$\lim_{n\to\infty} \frac{1}{n} \log_2 \sum_{w\in\mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \tag{48}$$

$$\lim_{n\to\infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \tag{49}$$

$$\dim_H^P \mathcal{N} = \delta. \tag{50}$$

If $B_\delta h = h$ with $h_s > 0$ for every state s, then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \tag{51}$$

is a probability law on surviving paths. For every $w = e_1 \cdots e_n \in \mathcal{W}_n$, define its probability under this law by

$$Q_\delta[w] = \prod_{i=1}^{n} Q_\delta(e_i \mid s_{i-1}).$$

Then it satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1-\delta)\log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \tag{52}$$

For a computable system with computable Q_δ, every nonterminating tape satisfies $\mathrm{Dim}_{\mathrm{eff},P}(\omega) \le \delta$. A tape that is Martin–Löf random under Q_δ satisfies $\dim_{\mathrm{eff},P}(\omega) = \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \delta$.

Proof. Since the graph contains a directed cycle, $\rho(B_0) \ge 1$. Since B_1 is substochastic, $\rho(B_1) \le 1$. If $\theta' > \theta$, then every nonzero entry contributed by an edge is smaller in $B_{\theta'}$ than in B_θ, because $p(e) < 1$. Strict Perron comparison on the common strongly connected support therefore gives $\rho(B_{\theta'}) < \rho(B_\theta)$. Together with continuity, this proves existence and uniqueness of δ. The relevant matrix identity is

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^{\mathsf{T}} B_\theta^n \mathbf{1},$$

where e_{s_0} is the coordinate vector of the initial state and $\mathbf{1}$ is the all-ones column vector. Perron–Frobenius theory proves (48); setting $\theta = 1$ gives (49).

If $\theta > \delta$, then $\rho(B_\theta) < 1$, so the total θ-weight of the length-n surviving cylinders tends to zero. These cylinders cover $\mathcal{N}$, giving $\dim_H^P \mathcal{N} \le \delta$. At $\theta = \delta$, (51) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because h is positive on a finite state space, there is a constant $C < \infty$, independent of w, such that $Q_\delta[w] \le C P[w]^\delta$. The mass-distribution principle states that a probability measure satisfying this cylinder bound forces $\dim_H^P \mathcal{N} \ge \delta$. It therefore gives the required lower bound, proving (50) and (52) (Bowen, 1975; Mauldin and Williams, 1988).

For a computable system, coding under Q_δ gives, for every surviving prefix w of length n,

$$K(w) \le -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a Q_δ-Martin–Löf-random path (Lutz, 2003; Athreya et al., 2007; Simpson, 2015). ☐

Corollary A.2 (Survival and dimension)**.** *Assume that the finite-state computation has a nonempty nonterminating set, and put*

$$\ell_- = \min_e \log_2 \frac{1}{p(e)}, \qquad \ell_+ = \max_e \log_2 \frac{1}{p(e)}.$$

Then

$$\left[1 - \frac{\alpha}{\ell_-}\right]_+ \le \dim_H^P \mathcal{N} \le 1 - \frac{\alpha}{\ell_+}, \tag{53}$$

where $[x]_+ = \max\{x, 0\}$*. Under computability, the middle quantity also equals* $\sup_{\omega \in \mathcal{N}} \mathrm{Dim}_{\mathrm{eff},P}(\omega)$*. Both inequalities become equalities when all surviving source outcomes have the same probability.*

Proof. For a reachable cyclic component C and $0 \le \theta \le 1$, entrywise comparison gives

$$2^{(1-\theta)\ell_-} B_{1,C} \le B_{\theta,C} \le 2^{(1-\theta)\ell_+} B_{1,C}.$$

Let $\alpha_C = -\log_2 \rho(B_{1,C})$. At $\theta = \delta_C$, where $\rho(B_{\delta_C,C}) = 1$, monotonicity of the spectral radius yields

$$1 - \frac{\alpha_C}{\ell_-} \le \delta_C \le 1 - \frac{\alpha_C}{\ell_+}.$$

Now use $\alpha = \min_C \alpha_C$ and $\dim_H^P \mathcal{N} = \max_C \delta_C$. □

Under a uniform q-ary source, let $M = (M_{st})$, where M_{st} counts the source symbols that move s to t. Then $B_\theta = q^{-\theta} M$. If

$$\alpha = -\lim_{n \to \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \ge n\}$$

is the survival exponent, then

$$\dim_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Writing $h_{\mathrm{surv}} = \log_2 \rho(M)$ for the topological entropy, in bits per step, of the surviving path space gives

$$\alpha = \log_2 q - h_{\mathrm{surv}}, \qquad \dim_H \mathcal{N} = \frac{h_{\mathrm{surv}}}{\log_2 q}.$$

Thus α is the entropy deficit imposed by survival, not an entropy function. Equation (52) is the finite-state form of the comparison used in Theorem 3.2.

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